

(a) Two relations, one recording

Shared excerpts: $A = \text{dog}$, $B = \text{rooster}$

Full recording x



$W_+ : A \rightarrow B$

$W_- : B \rightarrow A$

$q_+ : \text{dog followed by rooster}$

$q_- : \text{rooster followed by dog}$

Both descriptions hold in x .

Same sounds. Different targets.

The other window is a negative.

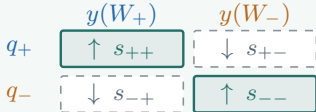
(b) Paired answer supervision

Full audio x + paired queries (q_+ , q_-)

Shared audio-language model
Frozen backbone; trainable LoRA

RelTwin candidate supervision

Training-only answer comparison



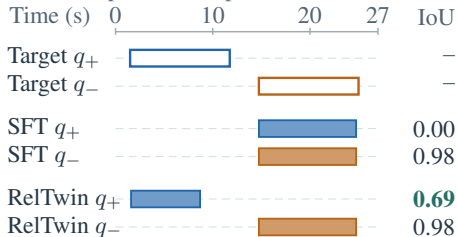
s : mean token log-likelihood

Row-wise candidate cross-entropy

$$\mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}$$

(c) Query-specific localization

Development example 10007-v0; seed 0



Same interval $\rightarrow$ query-specific targets

160 dev pairs, seed 0: SFT $\rightarrow$ RelTwin

Same-answer pairs: **37 $\rightarrow$ 1**

JointPairAcc: 53.13% $\rightarrow$ 84.38%

**Inference
(unchanged)**

Audio + query

Adapted timestamp generator

Predicted intervals

Candidate-free generation
Original inference pipeline